THE UNIVERSITY OF TEXAS AT AUSTIN
McCombs School of Business
Master of Science in Business Analytics

Project 1: CVaR Portfolio Optimization

Minimizing Downside Risk in an S&P 500 Portfolio

Course: Optimization for Decision Making
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Executive Summary

This report evaluates a Conditional Value-at-Risk (CVaR) optimization approach for building a low-risk portfolio from S&P 500 stocks, using 2025 data to construct portfolios and January–September 2026 data to test them.
Key findings:

- The CVaR model substantially reduces tail risk. The optimized portfolio ($\beta = 0.95$, R = 0.02%) had a CVaR of 0.93% in 2025, compared with 2.72% for the S&P 500 index. Out-of-sample in 2026, its CVaR rose to 1.16% but remained well below the index's 1.77%.
- A static portfolio does not preserve its risk profile. Out-of-sample CVaR was about 25% higher than in-sample, reflecting changing relationships between stocks over time. The allocation should not be held indefinitely.
- Protecting the worst month is possible, at a cost. Minimizing the maximum monthly CVaR cut the worst-month CVaR from about 4.06% to 1.23% in 2025, but produced a more concentrated portfolio (15 holdings, largest position 24.9%) and a higher full-year CVaR.
- Monthly reoptimization is not worthwhile. When both approaches were measured month by month in 2026, re-optimizing each month produced a slightly higher average monthly CVaR than keeping the 2025 portfolio (1.12% vs. 1.07%) and generated large month-to-month weight changes, with 140 violations of the 2-percentage-point stability limit.

Recommendation: Use the CVaR model as the core portfolio construction tool, but do not re-optimize every month. The allocation should be updated at a lower frequency (for example, once a year), since its tail risk increases as market conditions change, and any update should include stability constraints limiting each stock's weight change to 2 percentage points. If protection against a single severe month is the priority, the maximum-monthly-CVaR formulation is the better choice, provided the additional concentration is acceptable.

1. Introduction

In portfolio risk management, a key challenge is quantifying how much a portfolio could lose in a bad scenario, so that a manager can build portfolios that limit that downside. Value-at-Risk (VaR) has traditionally been used for this purpose: it represents the worst loss a portfolio is expected to experience over a given time horizon, at a chosen confidence level [2]. However, VaR has undesirable mathematical properties; it lacks subadditivity and convexity which makes it unreliable for combining risks across assets and impractical to use directly inside an optimization model.

Conditional Value-at-Risk (CVaR), also known as Expected Shortfall, addresses this limitation. It is defined as the average expected loss given that the VaR threshold has been exceeded [1]. Because CVaR is convex, it can be formulated as a linear program, making it a

practical and reliable choice for constructing an optimal, risk-minimizing portfolio which is why we use it instead of VaR in this project.

The objective of this analysis is to construct a portfolio of S&P 500 stocks that minimizes CVaR using 2025 data, evaluate how well that portfolio performs out-of-sample on 2026 data, and study how the choice of confidence level (β) and rebalancing frequency affect portfolio composition, risk, and stability.

2. Data

The stock universe for this analysis consists of the constituent companies of the S&P 500 index, obtained from the publicly available list on Wikipedia [3]. Daily adjusted closing prices ("Adj Close") for each stock were downloaded using the yfinance Python package.

Two separate periods of data were collected: calendar year 2025, used to construct the portfolio, and 2026 (from January 1 through September 14), used to evaluate the portfolio's out-of-sample performance. Daily returns for each stock were then calculated as the percentage change in price from one trading day to the next.

Of the 503 tickers in the S&P 500 universe, 5 stocks did not have complete adjusted-price histories over the training and evaluation periods. These stocks were excluded from the analysis, leaving 498 eligible stocks. Using a common set of stocks across both periods ensures that every asset considered during portfolio construction can also be evaluated consistently out-of-sample.

3. Problem Formulation

Decision variables:

- x_j: the weight (fraction of the portfolio) allocated to stock j, this is the main variable of interest, representing the portfolio itself.
- α (alpha): the threshold loss value that separates "normal" days from the worst-case days. This threshold corresponds to the portfolio's Value-at-Risk (VaR).
- u_k: the excess loss on day k beyond the threshold α. It equals zero if that day's loss did not exceed α and equals the amount by which it exceeded α otherwise.

Objective function:

Following the formulation of Rockafellar and Uryasev [4], the model minimizes Z, the portfolio's CVaR, which can be written in two equivalent ways.

Compact form:

$$Z = min_{x,\alpha,u}[\alpha + \frac{1}{(1-\beta)q}\sum_{k=1}^{q} u_k]$$

Explicit form (introducing Z as its own variable):

min Z

s.t.

$$Z = \alpha + \frac{1}{(1-\beta)q}\sum_{k=1}^{q} u_k$$

Both formulations are mathematically identical; the explicit form simply separates the objective value into its own variable.

Constraints:

$$\sum_j x_j = 1 \qquad \text{(weights sum to 1)}$$

$$x^T \bar{y} \geq R \qquad \text{(minimum expected return)}$$

$$u_k \geq -x^T \overline{y_k} - \alpha, \ \forall k \quad \text{(links } u_k \text{ to the portfolio's loss on day k)}$$

$$u_k \geq 0, \ \forall k \qquad \text{(excess loss cannot be negative)}$$

$$x_j \geq 0, \ \forall j \qquad \text{(non-negativity)}$$

where y_k is the vector of stock returns on day k, and $\bar{y}$ is the vector of average (mean) returns for each stock.

4. Base Case Results (β = 0.95, R = 0.02%)

4.1. Optimal Portfolio Composition

Solving the CVaR minimization model on 2025 return data produced a concentrated portfolio: only 31 of the 498 eligible stocks received a nonzero weight, while the remaining 467 were excluded entirely. Figure 1 shows these 31 holdings ranked by portfolio weight. The allocation is top-heavy, the five largest positions (Johnson & Johnson, 10.5%; Kroger, 8.1%; Altria, 7.5%; Consolidated Edison, 7.2%; and CME Group, 7.1%) together account for over 40% of the portfolio, while the smallest holdings (J.B. Hunt Transport Services, IDEX, American Electric Power, among others) each represent less than 0.5%. This concentration is consistent with the linear-programming structure of the model: without an explicit diversification constraint, the optimal solution can place substantial weight on a relatively small subset of stocks that minimizes the objective while satisfying the portfolio constraints.

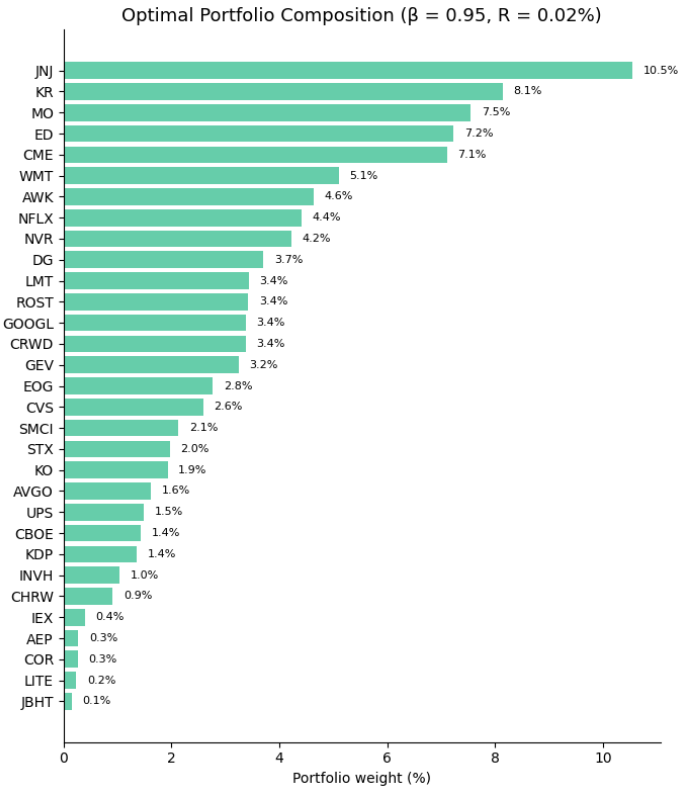


Figure 1. Optimal Portfolio Composition (β = 0.95, R = 0.02%)

4.2. In-Sample vs. Out-of-Sample CVaR

The portfolio optimized on 2025 data achieved an in-sample CVaR of 0.93% (β=0.95) on average, losses in the tail of the 2025 return distribution reached approximately 0.93% of portfolio value, with a corresponding VaR threshold of 0.62%. When the same portfolio weights were held fixed and re-evaluated against 2026 return data, the VaR threshold was re-estimated for the evaluation period at approximately 0.88%, and CVaR rose to 1.16%. This represents an increase of approximately 24.8% relative to the in-sample CVaR.

This degradation illustrates the non-stationary nature of financial return relationships: the correlations and tail behavior among stocks that the optimizer relied on to construct the 2025 portfolio do not hold exactly in the following period. A portfolio calibrated to minimize risk under one set of return data can therefore experience a different level of tail risk when market conditions change, even though its portfolio weights remain fixed.

4.3. Comparison Against the S&P 500 Index

To contextualize these results, the same CVaR methodology was applied to the S&P 500 index, treated as a single fully invested asset. In 2025, the index's CVaR was 2.72%, approximately 2.9 times the optimized portfolio's in-sample CVaR of 0.93%. In 2026, the index's CVaR was 1.77%, while the portfolio's out-of-sample CVaR was 1.16%, making the index's tail risk

approximately 1.5 times that of the optimized portfolio. In both periods, the optimized allocation produced a lower CVaR than the broad-market benchmark.

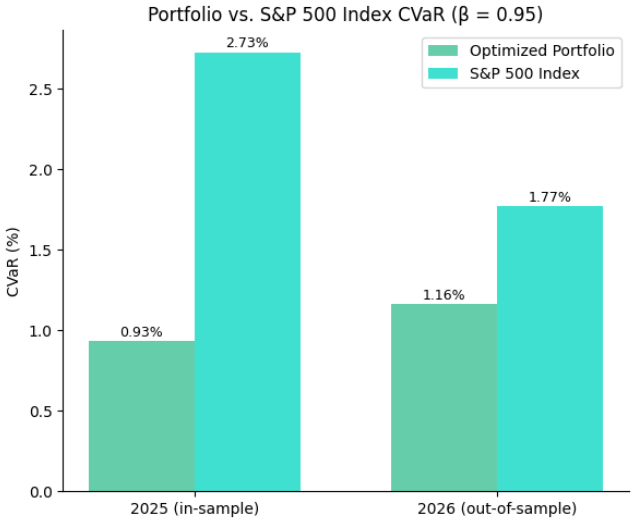


Figure 2. Portfolio vs. S&P 500 Index CVaR (β = 0.95)

4.4. Discussion: Should the Same Portfolio Be Kept?

Although the optimized portfolio continued to have a lower CVaR than the S&P 500 index out-of-sample, its own CVaR increased by approximately 24.8% from the training period to the evaluation period. The relative advantage over the index also narrowed, from approximately 2.9× in-sample to 1.5× out-of-sample. This gap between in-sample and out-of-sample CVaR suggests that a static allocation does not preserve the same risk profile when the underlying return relationships change. Therefore, keeping the same portfolio across different years is not advisable; the allocation should be re-evaluated as new data become available. The following analyses examine how the choice of β, a worst-month objective, and monthly reoptimization affect the resulting portfolio.

5. Sensitivity to Confidence Level (β)

5.1. Effect of β on Risk and Portfolio Allocation

To examine the effect of the confidence level, the portfolio model was rerun using β = 0.90, 0.95, and 0.99, while keeping the minimum expected return requirement unchanged. As β increases, the CVaR calculation focuses on a progressively more extreme portion of the loss distribution. The resulting VaR/CVaR values were 0.49%/0.75% at β = 0.90, 0.62%/0.93% at β = 0.95, and 1.23%/1.23% at β = 0.99. At β = 0.99, VaR and CVaR coincide because the tail

contains only about 2 to 3 of the 248 trading days in 2025, and the optimal portfolio's losses on those few days are essentially equal.

The portfolio allocation also changes materially with β. The β = 0.90 and β = 0.95 solutions each contain 31 holdings above the 0.1% reporting threshold, while the β = 0.99 portfolio contains only 17. The largest individual allocation increases from 12.7% at β = 0.90 and 10.5% at β = 0.95 to 25.1% at β = 0.99. Figure 3 illustrates how the optimal weights shift as the model places greater emphasis on more extreme tail losses.

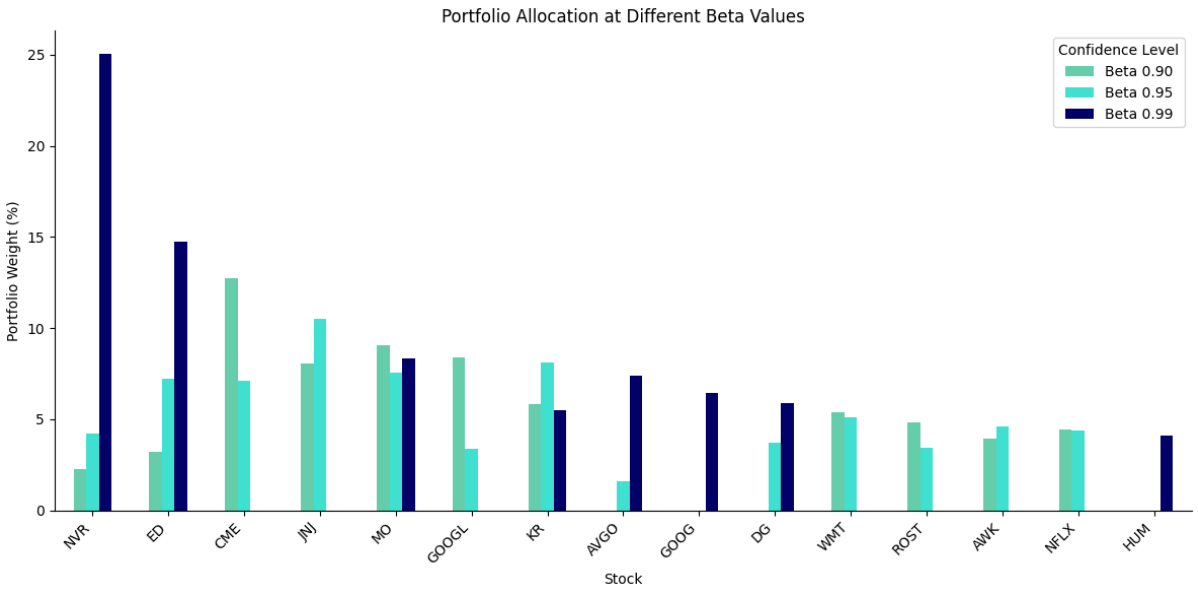


Figure 3. Portfolio Allocation at Different β Values

6. Maximum Monthly CVaR

6.1. Conservative Worst-Month Portfolio

The next model changes the objective from minimizing CVaR over the full annual return distribution to minimizing the maximum monthly 95% CVaR. This approach places greater emphasis on the single month with the largest tail loss rather than the overall annual risk measure. The optimized maximum monthly CVaR was 1.23%.

Changing the objective produced a substantially different portfolio. The base-case portfolio from Section 4 contains 31 holdings above the 0.1% threshold, with a largest individual weight of 10.5%. The maximum-monthly-CVaR portfolio contains only 15 holdings above the 0.1% threshold, and its largest allocation is 24.9% in NVR. MO, ED, COR, and HUBB also receive relatively large allocations. Figure 4 shows the resulting portfolio composition.

The increased concentration reflects the change in the optimization objective. Now, the portfolio is selected specifically to reduce the worst monthly CVaR rather than the CVaR of the entire annual return distribution. Because the two objective values measure different things, both portfolios were evaluated on both risk measures using the 2025 data.

The Section 4 portfolio achieved the lower full-year CVaR (0.93% vs. 1.23%), but its tail losses were concentrated in a single month: its CVaR in April 2025 reached approximately 4.06%. The maximum-monthly-CVaR portfolio limited its worst month to 1.23%, a reduction of 69.8%. April 2025 was the worst month for both portfolios, so the comparison directly shows how each approach performed under the same adverse market conditions. The conservative formulation therefore gives up some full-year tail performance in exchange for much better protection during the most adverse month, at the cost of a more concentrated portfolio.

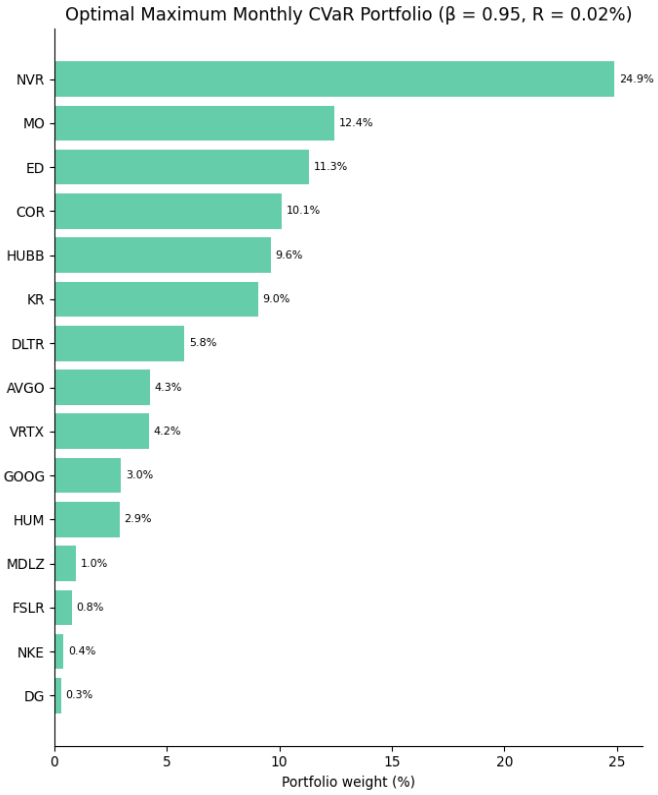


Figure 4. Optimal Maximum Monthly CVaR Portfolio (β=0.95, R=0.02%)

7. Monthly Portfolio Reoptimization

7.1. Out-of-Sample Monthly CVaR

The portfolio was re-optimized at the beginning of each month in the 2026 evaluation period. Each monthly portfolio was constructed using the previous one year of daily returns and was then evaluated using the actual daily returns observed during that month. This rolling procedure produced nine monthly portfolios from January through September 2026.

The realized monthly CVaR values ranged from 0.53% to 1.82%. The average monthly CVaR was 1.12%, with a standard deviation of approximately 0.48 percentage points. March and June produced the largest realized CVaR values, both approximately 1.8%, while February produced the minimum at approximately 0.53%. These results show that realized tail risk continued to vary considerably from month to month even when the portfolio was reoptimized regularly.

7.2. Comparison With the Static Portfolio

The out-of-sample CVaR of 1.16% reported in Section 4 is calculated over the full evaluation period, so it is not directly comparable with the monthly CVaR values above. To compare both approaches on the same basis, the static portfolio from Section 4 was also evaluated month by month in 2026. The static portfolio produced an average monthly CVaR of 1.07%, with a standard deviation of approximately 0.38 percentage points, and monthly values ranging from 0.69% to 1.89%. By comparison, the reoptimized portfolios averaged 1.12%, with a standard deviation of approximately 0.48 percentage points. The largest gaps occurred in May and June, when the reoptimized, portfolios reached a CVaR of approximately 1.46% and 1.81%, compared with 0.88% and 1.48% for the static portfolio.

Monthly reoptimization produced a lower CVaR than the static portfolio in 4 of the 9 months and slightly reduced the worst month (1.82% vs. 1.89%), but its average monthly CVaR was higher, and its results were more variable. In January, both approaches used the same portfolio, since the first reoptimized, portfolio is built from the same 2025 data. Based on these results, reoptimizing every month is not worthwhile in this sample: it does not reduce out-of-sample tail risk on average, and, as shown in Section 8, it produces large changes in the portfolio from one month to the next.

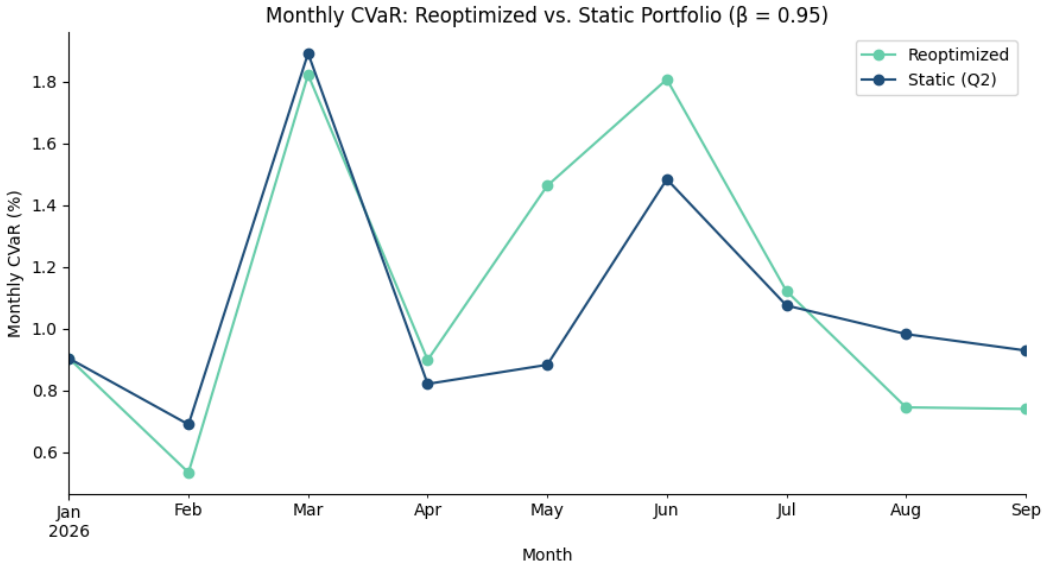


Figure 5. Monthly CVaR of the Reoptimized and Static Portfolios, 2026 (β = 0.95)

8. Portfolio Stability

The monthly portfolio sequence does not satisfy the assignment's stability requirement. A stable sequence requires every individual stock's weight to change by no more than 0.02 between consecutive months. The largest observed weight change was approximately 0.1417, or 14.17 percentage points, and 140 individual stock-month changes exceeded the 0.02 threshold.

This result shows that unconstrained monthly CVaR reoptimization can produce substantial changes in portfolio composition as the trailing return data changes. To enforce stability, two additional linear constraints could be imposed for each stock *j* in each month *t*:

$$x_{j,t} - x_{j,t-1} \leq 0.02$$

$$x_{j,t-1} - x_{j,t} \leq 0.02$$

Together, these constraints limit the absolute change in each stock's portfolio weight to two percentage points from one month to the next. The form of the CVaR objective remains unchanged, but the additional constraints reduce the feasible set and can therefore change both the optimal portfolio and the minimum attainable CVaR.

9. Recommendation

The CVaR optimization framework produced substantially lower measured tail risk than the S&P 500 benchmark in both the training and evaluation periods. However, the increase in out-of-sample CVaR from 0.93% to 1.16% shows that a portfolio optimized on one period does not maintain exactly the same risk profile when return relationships change.

The sensitivity analysis also shows that the choice of β materially affects portfolio composition, while minimizing the maximum monthly CVaR produces an even more concentrated allocation but cuts the worst-month CVaR by 69.8%. When measured month by month, monthly reoptimization did not reduce average out-of-sample CVaR (1.12% compared with 1.07% for the static portfolio), and it also produced an unstable sequence of portfolios, including 140 violations of the two-percentage-point stability criterion.

Based on these results, the base CVaR model provides a useful framework for constructing a low-tail-risk portfolio, but portfolio weights should not be treated as permanently optimal. Reoptimizing every month is not recommended; instead, the portfolio should be updated at a lower frequency, and any update should include the stability constraints described in Section 8 to limit large changes in individual positions. The maximum-monthly-CVaR formulation may

also be appropriate when protection against the single worst monthly tail outcome is the primary objective, although the resulting portfolio is substantially more concentrated.

10. Appendix A. Selected Code

```
TICKER_FILE = "TemplateTickerFile.csv"

# Training and evaluation dates
TRAIN_START = "2025-01-01"
TRAIN_END   = "2025-12-31"
EVAL_START  = "2026-01-01"
EVAL_END    = "2026-09-14"

# Benchmark required in Question 2b
BENCHMARK_TICKER = "^GSPC"

# Read the supplied ticker list dynamically
tickers_df = pd.read_csv(TICKER_FILE)
tickers = tickers_df.iloc[:, 0].dropna().astype(str).str.strip().tolist()
```

Code Excerpt 1. Generalized Input Configuration

```
# Confidence level for the CVaR (beta = 0.95 means we focus on the worst 5% of days)
beta=0.95

# Objective function: minimize alpha + the average excess loss over the worst days
profMod.setObjective(
    alpha + (1 / ((1 - beta) * q)) * gp.quicksum(u[k] for k in range(q)),
    GRB.MINIMIZE
)
```

```
# Minimum required average daily return (R = 0.02%)
R=0.0002

# Mean return of each stock, calculated from the 2025 daily returns
mean_returns=returns_2025.mean().values

# Constraint 1: portfolio weights must sum to 1 (fully invested)
profMod.addConstr(gp.quicksum(x[j] for j in range(n)) == 1, name="budget")

# Constraint 2: the portfolio's expected return must be at least R
profMod.addConstr(
    gp.quicksum(x[j] * mean_returns[j] for j in range(n)) >= R,
    name="min_return"
)

# Constraint 3: link u_k to the portfolio's loss on day k (the "dam" constraint)
# Convert returns to a plain array so we can index it by day (k) and stock (j)
returns_array = returns_2025.values

for k in range(q):
    portfolio_loss_k = -gp.quicksum(x[j] * returns_array[k, j] for j in range(n))
    profMod.addConstr(u[k] >= portfolio_loss_k - alpha, name=f"cvar_link_{k}")

# x_j >= 0 and u_k >= 0 were already set when we created those variables
# with lb=0, so no additional constraints are needed for those.
```

Code Excerpt 2. Base CVaR Objective and Constraints

```
# --- Part 2a: Evaluate the training-period portfolio out of sample ---
# The portfolio weights are fixed at their training-period optimal values.
# Alpha is re-solved for the evaluation return sample, as required by the
# project's F_beta(x) definition.

cvar_eval_fixed_weights, alpha_eval_for_fixed_x = evaluate_fixed_portfolio_cvar(
    optimal_weights,
    returns_2026,
    beta
)
```

Code Excerpt 3. Out-of-Sample CVaR Evaluation

```
# One maximum-CVaR variable
max_cvar = model.addVar(
    lb=-GRB.INFINITY,
    name="max_cvar"
)

# Objective
# Minimize the worst monthly CVaR
model.setObjective(max_cvar, GRB.MINIMIZE)
```

```
# max_cvar must be at least as large
# as the CVaR of this month
model.addConstr(
    max_cvar >= monthly_cvar
)
```

Code Excerpt 4. Maximum Monthly CVaR Formulation

```
for month in eval_months:

    # First day of the month being optimized
    month_start = month.to_timestamp()

    # Use the previous 12 months of return data
    training_start = month_start - pd.DateOffset(years=1)

    training_returns = returns_all[
        (returns_all.index >= training_start)
        & (returns_all.index < month_start)
    ]

    if training_returns.empty:
        raise ValueError(
            f"No trailing-year return data are available for {month}. "
            "Check the configured training and evaluation dates."
        )

    # Run the same CVaR optimization model used in Question 3
    weights, cvar, var = optimize_cvar_portfolio(
        training_returns,
        beta_value=beta,
        R_value=R
    )
```

Code Excerpt 5. Rolling One-Year Monthly Reoptimization

References:

[1] https://ryanoconnellfinance.com/expected-shortfall-cvar/
[2]https://medium.com/@simonleung5jobs/risk-management-in-financial-portfolio-710e270bf0e1
[3] https://en.wikipedia.org/wiki/List_of_S%26P_500_companies
[4] "Optimization of Conditional Value-at-Risk" by R. Tyrrell Rockafellar and Stanislav Uryasev.